

physiology

1- With respect to lung volumes ...

The volume of gas left in the lungs after a maximal expiration is the functional residual volume



Vital capacity is the volume exhaled when a maximal inspiration is followed by a maximal expiration

FRC can be measured directly with a spirometer

Helium dilution measures the inspiratory capacity

2- In which zone of the lung is there the blood flow during both of systole and diastole?



Zone-III

Zone-I

Zone-II

In all zones

3- What is the chloride shift in RBC during CO₂ transport?

Bicarbonate ion goes into the cell and chloride ion goes out of the cell



Chloride ion goes into the cell and bicarbonate ion goes out of the cell

Chloride ion goes out of the cell and hydrogen ion goes into the cell

Hydrogen ion goes into the cell and chloride ion goes out of the cell

4- With regard to the distribution of pulmonary blood flow, which one the following is correct?



The mean pulmonary arterial pressure is 15 mmHg

Hypoxia leads to pulmonary vasodilation

Blood flow is determined by the amount of alveolar ventilation

Typically there is a zone at the bottom that is not perfused

5- Which one the following factors causes that haemoglobin oxygen dissociation curve moves up and to the left?

Exercise

Acidosis



Hypothermia

Increased 2,3 DPG

6- Regarding ventilation/perfusion differences in the lungs ...

V/Q ratio is high at the bottom

In healthy individuals, anatomical dead space is more than physiological dead space

The relative change in blood flow from apex to bottom is less than relative change in ventilation

✓ V/Q differences are due to gravity

7- How long does it take for blood to pass through the pulmonary capillaries when cardiac output is normal?

✓ 0.8 s

0.3 s

1 s

2 s

8- In control of ventilation, the medullary chemoreceptors respond to ...

CO₂ tension

HCO₃⁻

✓ Hydrogen ion concentration

Oxygen tension

9- What is the effect of vagal reflex on the respiratory system?

Increasing the cilia beating frequency in the mucosiliary system

Bronchodilation

✓ Constriction of airways

Stimulation of beta receptors

10- Which one of the following is correct about the role of the muscles in inspiration?

Relaxation of the abdominal recti lengthens the chest cavity and leads to expiration

✓ Relaxation of diaphragm shortens the chest cavity and leads to expiration

Contraction of the scalene raises the rib cage and leads to expiration

Contraction of anterior serrati pulls the rib cage downward and leads to inspiration

11-

According to Laplace's law, how does the collapse pressure change as the alveolar radius is reduced by half (surface tension is constant)?

- Reduced to a quarter
- It is halved
- Increases to four times
- ✓ Doubles

12- With regard to the neural control of respiration ...

- The dorsal respiratory group has expiratory neurons
- The ventral respiratory group is located in the pons
- The main respiratory control centre is located in the pons
- ✓ Voluntary control system is located in the cerebral cortex

13- If a patient's blood carries 12 grams of hemoglobin per deciliter, what is the O₂ carrying capacity of his blood (milliliters per deciliter)?

- 18
- ✓ 16
- 14
- 12

14- Which one of the following factors leads to physiological dead space?

- Mixing pulmonary vein blood with bronchial vein blood
- A decrease in cardiac output to half
- A decrease in ventilation/perfusion ratio (V_A/Q) to zero
- ✓ An increase in V_A/Q ratio more than normal amount

15- Increase in which factor reduces the diffusion rate of oxygen?

- Solubility of gas
- ✓ Thickness of respiratory membrane
- Gas concentration gradient
- Surface area of respiratory membrane

16- Which of the following is correct about the ventral respiratory group (VRG) in the medulla?

Generates the inspiratory signal as a ramp signal

Its electrical stimulation only leads to an exhalation.

It has a fundamental role in respiratory basal rhythm

✓ It has a role in increasing pulmonary ventilation during muscle activity and exercise.

17- What is the normal P_{O_2} and P_{CO_2} in the venous blood, respectively?

95-40 mmHg

✓ 40-45 mmHg

40-95 mmHg

45-40 mmHg

18- compliance of the lung is reduced by all of the following except ...

By surgically removing part of the lung

alveolar edema

✓ emphysema

fibrosis

19- If the O_2 diffusing capacity of a patient is 15 ml/min/mmHg, what will be the CO_2 diffusing capacity (ml/min/mmHg)?

15

30

150

✓ 300

20- Trachea is a part of ...

Respiratory zone

✓ Anatomical dead space

Anatomical shunt

Alveolar dead space

21- Which one the following is correct about Surfactant?

✓ Prevents pulmonary edema

Surface tension is proportional to surfactant concentration

Reduces surface tension to zero

Is produced by type I alveolar cells

شرح پاسخ صحیح:

Anatomy

22-

Which of the following elements passes through the incisive canal?



Nasopalatin nerve

Superior labial artery

Sphenopalatin artery

Anterior ethmoidal nerve

23- Which bone is involved in both lateral wall and the roof of the nasal cavity?

Palatine



Ethmoid

Inferior concha nasalis

Maxillary

24-

All of the following paranasal sinuses open into the middle meatus....?

Middle ethmoidal sinus



Sphenoidal sinus

The anterior ethmoidal

maxillary sinus

25- The septum of the nose is made up all of the following bones except...?



Lacrimal

Ethmoid

Maxillary

Vomer

26- - All of the following arteries supplied nasal cavity except...?

Sphenopalatine artery

Anterior ethmoidal artery

Posterior ethmoidal artery



Superior laryngeal artery

27- Which one is adjacent to the mediastinal surface of the left lung?

Vena cava superior



Desending Aorta

Vena cava inferior

Arch of Azygous vein

28- All of the intrinsic muscles of larynx are supplied by recurrent laryngeal nerve except...?

Posterior cricoarytenoid

Oblique arytenoid



Cricothyroid

Vocalis

29- - Inferior border of the lung in the midclavicular line is at the level of?



Rib 8th

Rib 10th

Rib 6th

Rib 7th

30-

Which of the following elements pierces the thyrohyoid membrane?

9th Cranial nerve

11th Cranial nerve



Internal Laryngeal Nerve

Inferior Laryngeal artery

31- 5- Which of the following element is involved in forming the inferior quadriangular membrane?

Thyrohyoid membrane

Aryepiglottic fold



Vestibular ligament

Vocal fold

Embryology

32- Mature epithelial-endothelial (capillary) contact alveoli develop in which of the stages of lung maturation?

Saccular

Embryonic

Pseudoglandular



Alveolar

33- At the end of the sixth month, how many categories of branching divisions of bronchus are formed?



17

10

6

2

Histology

34- Which part of the lung tissue has a simple squamous cells?

Bronchiole

Terminal Bronchiole



Alveolar Duct

Bronchus

35- In which of the following is observed the Vocal Cord?

Olfactory Zone

Vestibuli Zone



Larynx

Respiratory Zone

36- Where is Hyaline Cartilage In Around of Burnouse?

Mucous

Nothing



Adventitia

Muscularis

37- What is the name Macrophage Alveolar Of Lunge ?

Pneumocyte IV



Pneumocyte III

Pneumocyte I

Pneumocyte II

38- In which of the following is observed the Unicell Gland Or Goblet Cell?

Alveolar Duct

Terminal Bronshiol

Vestibuli Zone



Theracea